Methodology for Deriving Internal Flux Linkage

- Assume the conductor carries a uniformly distributed current.
- Consider a thin cylindrical shell of radius x and thickness dx inside the conductor.
- Calculate the current enclosed within radius x.
- Apply Ampere's Circuital Law to determine the magnetic field intensity (H).
- Determine the magnetic flux density using: B = μH
- Calculate the elemental flux (dΦ) through the shell.
- Since only the enclosed current links this flux, determine the elemer
- Integrate dψ from the centre (x = 0) to the conductor surface (x = r) to obtain the total internal flux linkage.

